Equations of parallel and perpendicular lines

Core Concepts: Equations of parallel and perpendicular lines

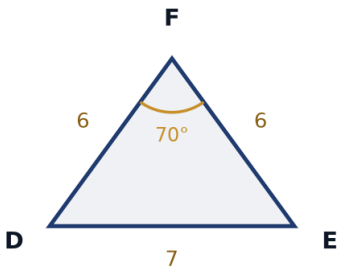
1 In $\triangle ABC$ studying equations of parallel and perpendicular lines, $m\angle A = 45^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 90°

b. 100°

c. 82°

d. 135°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length?

a. 6 cm

b. 8 cm

c. 12 cm

d. 16 cm

3 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 1$?

a. 4

b. 11

c. 9

d. 14

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying equations of parallel and perpendicular lines, $m\angle A = 45^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 83°

b. 93°

c. 75°

d. 135°

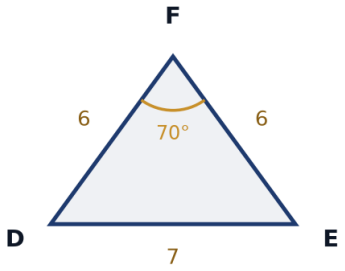
5 A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length?

a. 9 cm

b. 12 cm

c. 16 cm

d. 36 cm



6 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 2$?

a. 9

b. 16

c. 14

d. 19

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying equations of parallel and perpendicular lines, $m\angle A = 45^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 76°

b. 86°

c. 68°

d. 135°

8 A geometric figure scaled by factor $k = 4$ has an original side length of $4\ \text{cm}$. What is the corresponding image length?

a. 12 cm

b. 16 cm

c. 20 cm

d. 64 cm

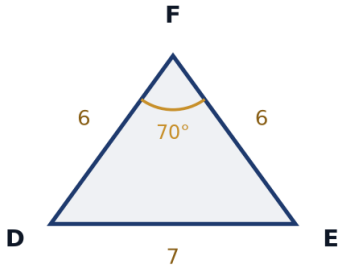
9 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$?

a. 14

b. 21

c. 19

d. 24



Exam-Style Problem Solving & Modeling

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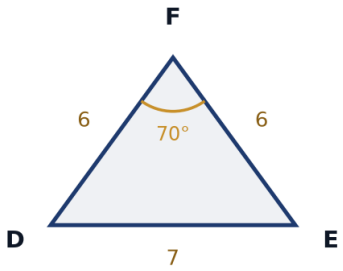
10 In $\triangle ABC$ studying equations of parallel and perpendicular lines, $m\angle A = 45^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 69°

b. 79°

c. 61°

d. 135°



11 A geometric figure scaled by factor $k = 5$ has an original side length of 4 cm. What is the corresponding image length?

a. 15 cm

b. 20 cm

c. 24 cm

d. 100 cm

12 On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$?

a. 19

b. 26

c. 24

d. 29

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